MIL-P-46131B
17 November 1975
SUPERSEDING
MIL-P-46131A(MR)
18 December 1974

MILITARY SPECIFICATION

PLASTIC MOLDING AND EXTRUSION MATERIAL, POLYPHENYLENE OXIDE, MODIFIED, GLASS FIBER REINFORCED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements of glass fiber filled modified polyphenylene oxide materials for use in moldings and extrusions (see 6.1).

1.2 Classification. The glass fiber filled modified polyphenylene oxide material shall be of the types and of the classes, as specified (see 6.2).

Type I - 20 percent glass fiber

Class 1 - General purpose
Class 2 - Lowest flammability

Type II - 30 percent glass fiber

Class 1 - General purpose
Class 2 - Lowest flammability

Type III - 40 percent glass fiber

Class 1 - General purpose

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 9330

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SPECIFICATIONS

FEDERAL
PPP-D-723 - Drums, Fiber
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-Gallon

MILITARY
MIL-P-46129 - Plastic Molding and Extrusion Material, Polyphenylene Oxide, Modified

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 149 - Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies
D 150 - A-C-Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials
D 256 - Impact Resistance of Plastics and Electrical Insulating Materials
D 257 - D-C Resistance or Conductance of Insulating Materials
D 570 - Water Absorption of Plastics
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing
D 635 - Flammability of Rigid Plastics Over 0.127 cm (0.050 incn) in Thickness
D 638 - Tensile Properties of Plastics
D 648 - Deflection Temperature of Plastics Under Load
D 792 - Specific Gravity and Density of Plastics by Displacement

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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Underwriters' Laboratories (UL), Inc.

UL 94 - Test for Flammability of Plastic Materials, dated December 1973.

(Copies may be purchased directly from Underwriters' Laboratories, Inc. 207 E. Ohio Street, Chicago, Illinois, 60611.)

National Motor Freight Traffic Association, Inc.,Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Preproduction testing. When specified by the procuring activity (see 6.2) preproduction testing shall be made on the first lot of material to be submitted for acceptance (see 4.2.3.2).

3.2 Material. The material shall consist of glass fiber filled modified polyphenylene oxide resin formulated for use in injection molding and extrusion. The modified polyphenylene oxide resin shall conform to MIL-P-46129. Nominal glass fiber content shall be as specified in 1.2 for types I, II, and III. Actual glass fiber content shall be ± 2 percent from nominal. The glass fiber shall be "E" glass (see 3.2.1)

3.2.1 "E" glass. The chemical composition of "E" glass, as percent by weight, shall be within the limits, as follows:

B_2O_3	5-10
CaO	16-25
Al_2O_3	12-16
SiO_2	52-56
MgO	0-5
Na_2O and K_2O	0-2
TiO_2	0-0.8
Fe_2O_3	0.05-0.4
F_2	0-1.0

3.3 Property values. Specimens prepared from the material, as specified in 4.3.1, shall conform to the property values specified in table I, when tested as specified in the applicable procedure of 4.3. When specified by the procuring activity (see 6.2), the electrical properties specified in table II shall be included as requirements.

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Table I. Property values

Property	Value				
	Type I		Type II		Type III
	Class 1	Class 2	Class 1	Class 2	Class 1
Specific gravity, min.	1.19	1.19	1.25	1.25	1.36
max.	1.23	1.23	1.29	1.29	1.40
Deflection temperature under load, 1/4 inch thick, 264 psi., °C, min.	132	132	151	146	154
°F, min.	270	270	305	295	310
Impact strength, Izod, 1/4 inch thick, ft. lbs. per inch of notch, min.	1.6	1.6	1.9	1.9	1.7
Tensile yield strength, psi, min.	14,000	14,000	16,000	16,000	18,000
Modulus of elasticity, psi, min.	750,000	750,000	1,050,000	1,050,000	1,150,000
Water absorption, percent, max.	.16	.16	.14	.14	0.10
Flammability, max. by 4.3.8.1	Less than 4 inches[1/]	NA[2/]	Less than 4 inches[1/]	NA[2/]	Less than 4 inches[1/]
by 4.3.8.2	NA[2/]	94V-1[3/]	NA[2/]	94V-1[3/]	NA[2/]

1/ Burning extent of less than 4 inches by ASTM test specified in 4.3.8.1.
2/ NA - Not applicable.
3/ 94V-1 by the Underwriters' Laboratories test specified in 4.3.8.2.

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Table II. Electrical property values[1/]

Property	Value				
	Type I		Type II		Type III
	Class 1	Class 2	Class 1	Class 2	Class 1
Dielectric constant at 1 megahertz max.	3.0	3.1	3.0	3.0	3.0
Dissipation factor at 1 megahertz max.	0.003	0.003	0.003	0.003	0.003
Dielectric strength, short time test, volts per mil, min.					
1/8 inch thick specimen	500	500	500	500	500
1/32 inch thick specimen	-	1000	-	-	-
Volume resistivity, ohm-cm, min.	-	10^{16}	-	-	-

[1/] Applicable only when specified by the procuring activity (see 6.2).

3.4 Form. The material shall be furnished in the form of molding powder, granules or pellets, as specified (see 6.2). When a particular particle size is specified, the limits and method of testing shall be as specified by the procuring activity (see 6.2 and 6.3).

3.5 Color. The material shall be of the color and opacity specified (see 6.2 and 6.3). When close color limits are specified, color matching shall be performed on molded plaques of the thickness specified, in accordance with the method specified by the procuring activity.

3.6 Uniformity. The powder, granules or pellets shall be uniform in color within each container and from container to container. The form shall be uniform from container to container. The material shall be free from contamination as determined by visual examiniation.

3.7 Suitability (compatibility) for use with explosives. (Applicable only when specifically designated in the applicable contract, order, drawing or specification). When so specified, application shall be made to a Government laboratory or other Government approved laboratory designated by the procuring activity for determination or suitability of the material for use with a particular explosive or explosives (see 6.2 and 6.4).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same class submitted for inspection at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, inspected, and tested in accordance with the requirements of this specification and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification shall govern. A supplier's certificate of compliance with the requirements specified in 3.2, and the actual glass fiber content shall be furnished.

4.2.2 Inspection of material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 50 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination specified in table III shall be approximately one pound.

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Table III. Examination of the material for defects in appearance and workmanship

Examine	Defect
Appearance and workmanship	Form not as specified
	Color not as specified
	Form not uniform
	Color not uniform
	Not clean, presence of foreign material

4.2.2.1.2 Examination of the preparation for delivery requirements. An examination shall be made in accordance with table IV, to determine that packaging, packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

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Table IV. Examination of the preparation for delivery

Examine	Defect
Packing	Not level specified; not in accordance with contract requirements
	Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability
	Inadequate application of components such as: incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers
Quantity of material	Less than specified or indicated quantity
Weight	Gross weight exceeds specified requirements
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete, of improper size, location, sequence, method of application, or not in accordance with contract requirements.

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQLs) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

4.2.3 Classification of testing. All tests shall be classified as follows:

(a) Lot acceptance tests(see 4.2.3.1).
(b) Preproduction tests (see 4.2.3.2).
(c) Electrical property tests (see 4.2.3.3).

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4.2.3.1 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and, in conjunction with the above examination, shall be the basis for acceptance or rejection of the lot. The lot acceptance tests shall consist of the tests for all properties shown in table V. When no preproduction testing has been required (see 4.2.3.2), lot acceptance tests for the first lot only shall include impact strength and water absorption tests in addition to those specified in table V.

Table V. Lot acceptance tests for types I, II, and III

Test	Applicable to class	Test method paragraph
Specific gravity	1 & 2	4.3.3
Deflection temperature	1 & 2	4.3.4
Impact strength	1 & 2	4.3.5
Tensile yield stress	1 & 2	4.3.6
Flammability	2 only	4.3.8.2 1/

1/ Unless otherwise specified (see 6.2)

4.2.3.2 Preproduction tests. When required by the procuring activity (see 6.2), preproduction tests shall be made on the first lot of material furnished under this specification and on any other lot specified, and shall consist of tests for all properties specified in table I. When electrical property values are required, preproduction tests shall include tests for all properties specified in tables I and II.

4.2.3.3 Electrical property tests. Electrical property tests shall be made on any lot specified by the procuring agency (see 6.2), and shall be included in the basis for acceptance or rejection of the lot.

4.2.4 Testing. The material shall be tested for the applicable characteristics listed in table I, in accordance with the test methods specified herein. The lot size for the purpose of determining the sample size (see MIL-STD-105) for testing shall be expressed in units of 200 pounds of material. The sample unit shall consist of sufficient material to prepare all specimens required for testing. The inspection level shall be S-1, and test results for each property and each unit of product shall meet requirements. Test results shall be averages of values obtained from all specimens for each property tested per unit of product, except for the impact strength determination (see 4.3.5). Each test specifies the number of specimens to be tested for each unit of product in the sample size.

4.3 Test methods.

4.3.1 Preparation of specimens. Material in the form of pellets shall be used as such for the melt viscosity test. Material in the form of powder shall be injection molded into a massive form, such as tensile specimens, and then cut up into small pieces for use in the melt viscosity test. For all other tests, the material shall be injection molded into specimens as described in the individual test method using the technique of molding recommended by the supplier. In the absence of such instructions, the following procedure shall be used:

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(a) Dry material for 2 hours at 110° ± 2°C (230° ± 3.6°F).
(b) Use a screw type injection molding machine under the following conditions:

(1) Melt temperature, 260 to 288°C (530-570°F)
(2) Mold temperature, 82 to 99°C (190-220°F)
(3) Cycle, seconds, minimum: 20
(4) Pressure: Sufficient to give 3 seconds or less injection time.

4.3.2 Conditioning. All specimens, except those for melt viscosity, water absorption and flammability, shall be conditioned in accordance with procedure A of ASTM D 618. The specimens for melt viscosity need only be sufficiently dry that moisture from them will not interfere with the test. When necessary, the drying procedure of 4.3.1 shall be used. The specimens for water absorption and flammability shall be dry-as-molded, except that half of the dry-as-molded specimens for use in 4.3.8.2 shall be subjected to the additional conditioning specified therein. (The "as received" referenced in 4.3.8.2 shall be interpreted as "dry-as-molded".)

4.3.3 Specific gravity. A minimum of 2 molded specimens of any convenient size shall be tested in accordance with method A-1 or A-2 of ASTM D 792.

4.3.4 Deflection temperature under load. A minimum of two 1/4-inch thick specimens shall be tested in accordance with procedure A of ASTM D 648.

4.3.5 Impact strength. A minimum of 5 specimens, each 1/4-inch thick shall be tested in accordance with method A of ASTM D 256. The median value, in foot pounds per inch of notch, shall be reported.

4.3.6 Tensile yield strength and modulus of elasticity. Five specimens shall be tested for tensile yield strength and modulus of elasticity in accordance with ASTM method D 638. The specimens shall be 1/8-inch thick and shall conform to the dimensions of type I of Figure 1 of that method. Speed B shall be used. The yield strength is the stress at the first point at which the load vs extension curve becomes horizontal.

4.3.7 Water absorption. Three specimens shall be tested in accordance with ASTM D 570, using options 5(b) and 6(a), but without reconditioning (section 7).

4.3.8 Flammability.

4.3.8.1 ASTM D 635 (for class 1 material). A minimum of 10 specimens shall be tested in accordance with ASTM D 635. The specimens shall have nominal dimensions of 5 inches by 1/2 inch by 1/4 inch.

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4.3.8.2 Underwriter's test (for class 2 material). Twelve specimens (6 of each of two different thicknesses, 3 of each thickness being subjected to one of two different conditioning procedures) shall be tested in accordance with the Underwriter's Laboratories UL 94, dated December 1973. All specimens shall meet the flammability requirement specified in table I.

4.3.9 Dielectric constant and dissipation factor. The dielectric constant and dissipation factor shall be determined on 5 specimens, each 4 inches in diameter and 1/8 inch thick, in accordance with ASTM D 150 using a frequency of one megahertz.

4.3.10 Dielectric strength. The dielectric strength of 5 specimens each 4 inches in diameter and 1/8 inch thick, shall be determined by the "short-time test" of ASTM D 149. When specified by the procuring activity (see 6.2), thickness shall be 1/32 inch for type I, class 2 material (see 6.2).

4.3.11 Volume resistivity (applicable to type I, class 2 material only. Four specimens, each 2 inches in diameter by 1/8 inch thick shall be tested for volume resistivity in accordance with ASTM D 257.

5. PREPARATION FOR DELIVERY

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, the material shall be packed in one of the following types of containers:

a. Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A, class 1, in quantities of 200 pounds, maximum.

b. Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified, the material shall be packed in one of the following types of containers:

a. Fiber drums conforming to PPP-D-723, type I, grade A, class 1, in quantities of 200 pounds maximum.

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b. Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds, maximum.

Insofar as practical, drums shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Drums shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with a 0.004 inch thick polyethylene liner properly heat sealed.

5.1.3 Level C. The material shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking. In addition to any special marking specified in the contract or order (see 6.2) containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Material conforming to this specification is recommended for injection molding or extruding into items with good mechanical properties, good electrical properties in high humidity environments, fungus resistance, hydrolytic stability, and good strength at elevated temperatures. This material has a maximum continuous use temperature at 100°C (212°F), at which temperature it retains more than half of its room temperature strength and stiffness. Exposure at temperatures up to its deflection temperature 121°C (250°F) are feasible for short times in air or for longer times in non-oxidizing environments. The material is resistant to most aqueous environments, but is swollen, dissolved, or stress cracked under some conditions in some chlorinated or aromatic organic environments. Specific end use testing is recommended to establish the effects of time, temperature, and environment on the molded item. Some typical applications include connectors which require electrical properties, and surgical instruments which require strength and stiffness plus ability to resist repeated steam sterilizations. Class 2 material is recommended for handles of electrical power tools and similar applications requiring extremely low flammability.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. Preproduction inspection, if required (see 3.1).
d. Electrical property tests, if required (see 3.3 and table II).
e. Thickness of specimen for dielectric strength (see table II and 4.3.10), for type I, class 2 material when 1/32 inch is required
f. Particle shape, size range and method of determination, if required (see 3.4 and 6.3).
g. Color and opacity required, and method of determination if required (see 3.5 and 6.3).
h. Resistance to explosives, if required (see 3.7 and 6.4).
i. Level of packing required (see 5.1).
j. Special marking, if required (see 5.2).

6.3 Determination of form. When a particular form, particle size or color is required, a standard sample, accompanied by borderline standards or by a written statement defining the permissible tolerances and methods of determination, may be furnished by the procuring activity or furnished by the supplier and approved by the procuring activity. In the case of color, one standard may be required for use in judging the uniformity of the color of the powder or pellets under 3.4, while another standard may be required for judging the color of a plaque molded from the material.

6.4 Suitability for use with explosives. Information concerning the suitability of many plastics for use with various explosives and chemicals under various conditions is on file at Picatinny Arsenal, Dover, New Jersey. Procuring activities desiring information on this subject should first contact Picatinny Arsenal to determine whether the information is already available.

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User activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: ____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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